Most Used & Asked Topics in React/Next.js

Core React Concepts

1. Components & Props

- Functional vs Class components
- Props drilling and prop types
- Children props
- Component composition

2. React Hooks (Most Important)

- `useState` - state management
- `useEffect` - side effects, lifecycle
- `useContext` - context API
- `useRef` - DOM references, persisting values
- `useMemo` - memoization
- `useCallback` - function memoization
- `useReducer` - complex state logic
- Custom hooks

3. State Management

- Local state vs Global state
- Context API
- **Redux** (Redux Toolkit)
- **Zustand** (lightweight)
- **React Query/TanStack Query** (server state)

4. Component Patterns

- Controlled vs Uncontrolled components
- Higher-Order Components (HOC)
- Render Props
- Compound Components
- Container/Presentational pattern

5. React Router

- Route configuration
- Navigation (Link, useNavigate)
- Dynamic routing
- Protected routes
- URL parameters

Next.js Specific Topics

6. Rendering Strategies

- **SSR** (Server-Side Rendering)
- **SSG** (Static Site Generation)
- **ISR** (Incremental Static Regeneration)
- **CSR** (Client-Side Rendering)
- `getServerSideProps`, `getStaticProps`, `getStaticPaths`

7. Next.js App Router (v13+)

- File-based routing
- Server Components vs Client Components
- `use client` directive
- Layout and templates
- Loading & error handling
- Streaming and Suspense

8. Next.js Features

- API Routes
- Image optimization (`next/image`)
- Font optimization (`next/font`)
- Metadata API (SEO)
- Middleware
- Environment variables

9. Data Fetching

- `fetch` API
- SWR (Stale-While-Revalidate)
- React Query/TanStack Query
- Axios integration
- Error handling & loading states

CSS & Styling

10. Modern CSS Approaches

- **Tailwind CSS** (highly used)
- CSS Modules
- Styled Components
- CSS-in-JS
- Responsive design (mobile-first)
- Flexbox & Grid

Performance Optimization

11. React Performance

- Code splitting (`React.lazy`, `Suspense`)
- Memoization (`React.memo`, `useMemo`, `useCallback`)
- Virtual DOM concepts

- Bundle size optimization
- Lazy loading images

12. Next.js Performance

- Dynamic imports
- Image optimization
- Font optimization
- Bundle analysis
- Caching strategies

Form Handling

13. Form Libraries

- React Hook Form (most popular)
- Formik
- Validation (Zod, Yup)
- Controlled forms

TypeScript

14. TypeScript with React

- Component typing
- Props interface
- Event types
- Generic components
- Type vs Interface

Testing

15. Testing Libraries

- Jest
- React Testing Library

- Cypress/Playwright (E2E)

Build Tools & Deployment

16. Development Tools

- Vite
- Webpack basics
- ESLint & Prettier
- Git workflows

Job Preparation Plan for 6sense HQ Limited

Week 1-2: Core Fundamentals

JavaScript ES6+

- Arrow functions, destructuring
- Promises, async/await
- Array methods (map, filter, reduce)
- Spread/rest operators
- Modules (import/export)

React Essentials

- Build 2-3 small projects with hooks
- Practice useState, useEffect deeply
- Understand component lifecycle
- Props and state management

Week 3-4: Next.js & Advanced React

Next.js Basics

- Create a Next.js project
- Implement different rendering methods
- Build API routes
- Image optimization practice

State Management

- **React Query** (Priority - mentioned in job)
- Redux Toolkit basics
- Zustand (lightweight alternative)

Week 5-6: Production Skills

API Integration

- RESTful API consumption
- Error handling
- Loading states
- Authentication flows

Responsive UI

- Build pixel-perfect components from Figma
- Mobile-first approach
- Tailwind CSS mastery
- Cross-browser compatibility

Git & Collaboration

- Git branching strategies
- Pull requests
- Code review practices
- Merge conflict resolution

Interview Preparation

Technical Topics to Master

1. React Interview Questions:

- Virtual DOM explanation
- Reconciliation process
- useEffect cleanup
- Keys in lists
- Controlled components
- Event handling
- Performance optimization

2. Next.js Questions:

- SSR vs SSG vs CSR differences
- When to use each rendering method
- API routes use cases
- Image optimization benefits
- Routing system

3. JavaScript Questions:

- Closures
- Event loop
- Prototypal inheritance
- `this` keyword
- Promises vs async/await

4. Coding Challenges:

- Array manipulation
- String problems
- Object operations
- DOM manipulation
- Implement common components (accordion, modal, tabs)

Projects to Build

1. E-commerce Product Listing

- Next.js with API integration
- React Query for data fetching
- Filters and search
- Responsive design
- Shopping cart (Zustand/Context)

2. Dashboard Application

- Charts/data visualization
- CRUD operations
- Authentication
- Form handling (React Hook Form)
- Table with pagination

3. Blog/Portfolio

- Next.js SSG
- Markdown support
- SEO optimization
- Dark mode toggle

Day Before Interview

✅ Review your projects thoroughly

✅ Prepare questions about their tech stack

✅ Review common React/Next.js patterns

✅ Practice explaining your code decisions

✅ Prepare examples of problem-solving

Key Strengths to Highlight

- Production-level project experience
- Clean, maintainable code writing ability

- API integration experience
- State management knowledge
- Responsive UI development
- Git workflow proficiency
- Learning agility with modern tools

Focus Areas Based on Job Requirements:

1. **React Query** - They specifically mentioned this
2. **Next.js** - Preferred, so prioritize
3. **Component architecture** - Clean, reusable components
4. **API integration** - Practice extensively
5. **Responsive UI** - Pixel-perfect implementation